A-A-53979
15 September 1989

COMMERCIAL ITEM DESCRIPTION

WIRE, NONELECTRICAL (Platinum Alloy)

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-W-3662B which is cancelled.

Salient characteristics:

This commercial item description covers platinum alloy wire suitable for chemical and serological laboratory use in the following grades:

Grade A. - Platinum - Ruthenium, Alloy Wire

Grade B. - Platinum - Iridium, Alloy Wire

Material. Materials shall be of the highest quality and entirely suitable for the purpose intended.

Chemical composition. Grade A wire shall be 95% platinum and 5% ruthenium. Grade B wire shall be 85% platinum and 15% iridium. Metals used in the alloys shall be not less than 99.5% pure. A tolerance of minus 0.5% on individual alloy constituents is permissable.

Physical properties. The physical properties of the wires shall conform to requirements of table I.

TABLE I. Physical properties

Grade	Melting point	Tensile strength	Weight per wire
A	1800 $\pm$ 10^0 C	100,000 to 135,000 psi	0.999 grams, min
B	1820 $\pm$ 10^0 C	100,000 to 135,000 psi	0.425 grams, min

Dimensions and tolerances. The dimensions and tolerances shall be as specified in table II.

AMSC N/A FSC 9545

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:27Z
Check the source to verify that this is the current version before use.

TABLE II. Dimensions and tolerances

Grade	Gauge (Brown & Sharpe)	Diameter and tolerance (inch)	Length (inches)
A	22	0.0254 ± .0002	6 ± 1/32
B	26	0.0159 ± .0002	6 ± 1/32

Finish. The wire shall have a bright smooth finish.

Workmanship. The wire shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One six inch length of wire, of the grade specified, constitutes one unit of issue. Each unit of issue shall be packaged straight, enclosed in a suitable envelope. Envelope closure shall be adequately secured to prevent spilling of contents during normal handling.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Source: https://assist.dla.mil -- Downloaded: 202640-26T09:27Z
Check the source to verify that this is the currentversion before use.

Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

Sampling for examination. The wire shall be examined for defects including, but not limited to, those listed below. The unit of product for examination purposes shall be one wire. Sampling for examination shall be in accordance with MIL-STD-105 at inspection level II.

Defects

Diameter of wire not as specified.
Diameter of wire not uniform.
Finish not smooth and bright.
Cracks, laminations, pitted areas present.
Length of wire not as specified.

Sampling for tests. A sufficient quantity of wire from each grade (both A and B), from each lot shall be selected for the chemical analysis test. Two units of wire per lot shall be subjected to the melting point test. Five units of wire from each lot shall be subjected to the ultimate tensile strength and individual weight tests. For the ultimate tensile strength test, the average of the 5 specimens shall be reported and used for determining compliance with the requirements of this document. For the melting point and individual weight test, a lot shall be rejected if any of the samples fails the test.

Tests. Tests shall be conducted to determine compliance with specification requirements.

Chemical analysis. The samples selected shall be analyzed in conformance with methods 111 or 112 of Federal Standard No. 151. In case of a dispute, method 111 shall be used for referee analysis.

Melting point. Melting point shall be determined by optical pyrometer, heating-cooling curve, or other method acceptable to the Government.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:27Z
Check the source to verify that this is the current version before use.

Ultimate tensile strength. Determinations for ultimate tensile strength shall be made on 5 specimens selected from each lot or control number. The load shall be applied slowly and continuously at a loading rate of 1 inch per minute until rupture occurs. The average of the 5 specimens shall be reported and used for determining compliance with specification requirements.

Weight. Weighing of individual units of issue and of each delivery to the Government shall be performed on a precision balance accurate to 0.001g for individual weighing and accurate to 0.01g for each delivery to the Government.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pounds units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:27Z
Check the source to verify that this is the current version before use.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

National Stock Number	Item Identification
9545-00-448-9010	WIRE, NONELECTRICAL, Platinum Alloy, 22 Gage, 6 inch
9545-00-448-9110	WIRE, NONELECTRICAL, Platinum Alloy, 26 Gage, 6 inch

Ordering data. Intermediate/exterior package quantities, labeling and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:27Z
Check the source to verify that this is the current version before use.

A-A-53979

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

PROJECT NO. 9545-0005
LOCATION: NCR DISK 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:27Z
Check the source to verify that this is the current version before use.